SEC P vs NP log 2026-05-01

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 08:00 UTC

SEC P vs NP — daily log 2026-05-01

35 cycles recorded

Spectral Norm of SOS Relaxation and Refutation Size in Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × Sum-of-Squares Refutation Size
- **Recorded**: 2026-05-01 00:11 UTC
- **Entry ID**: `56039c424522`

Statement

For random 3-SAT instances with `n` variables, the spectral norm of the SoS relaxation matrix is Θ(√(log n)), and the size of the SoS refutation is Θ(n^{2/3} / √(log n))

Rationale

The spectral properties of the SoS relaxation matrix encode geometric constraints on the feasible region, which directly influence the hierarchy's ability to find contradictions. Random matrix theory provides tools to analyze these constraints in high-dimensional random CSPs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `4.74s`


```
tation_size', 'metric_value': 6.0896560294980295, 'instances_tested': 30, 'conjecture_holds': True, 
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 6.0896560294980295, 'instances_tested': 30
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.058854520080662, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.058854520080662, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 2.3048511805828995, 'instances_tested': 30
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.6960044369755285, 'instances_tested': 30
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.6960044369755285, 'instances_tested': 30
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 6.0896560294980295, 'instances_tested': 30
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 5.235179597267003, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.058854520080662, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 5.235179597267003, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.058854520080662, 'instances_tested': 30,
TRIAL: {'metric_name': 'refutation_size', 'metric_value': 3.058854520080662, 'instances_tested': 30,
```

```
RESULT: SUPPORTED mean=4.090674347243454 std=1.3413521963407913 support_fraction=1.0
```

Judge reasoning

Test results show constant metric values across 30 instances, failing to demonstrate asymptotic scaling with n. While all trials support the conjecture, the critic correctly challenges the validity of scaling claims with insufficient n. | next: Test with larger n (e.g., n=10^4) to observe asymptotic behavior of refutation_size metric

---

Spectral Radius of SOS Moment Matrix Bounds Max-CUT Approximation Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Moment Matrix
- **Recorded**: 2026-05-01 00:46 UTC
- **Entry ID**: `f7f5b0531d9e`

Statement

For any max-CUT instance on n vertices, the spectral radius $\rho(M)$ of its degree-d SOS moment matrix satisfies $\rho(M) = \Theta(\sqrt{n})$ if and only if the SOS relaxation achieves approximation ratio $\Theta(1/\sqrt{n})$.

Rationale

The spectral properties of SOS moment matrices encode the curvature of the underlying polynomial optimization problem. Bounding the spectral radius provides a handle on the trade-off between relaxation degree and approximation quality, leveraging real algebraic geometry's tools for polynomial positivity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0bc789b2.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0bc789b2.py", line 106, in run_trial
    rho_M = spectral_radius(M)
            ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0bc789b2.py", line 89, in spectral_radi
    x = [Fraction(1, math.sqrt(n))] * n
         ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

Test crashed due to TypeError in Fraction initialization, preventing data collection | next: Fix Fraction argument type conversion in test code to handle float inputs

---

Secant Variety Dimension Lower Bounds for Disjointness

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry × Communication Complexity
- **Recorded**: 2026-05-01 01:19 UTC
- **Entry ID**: `0fa75c46e35c`

Statement

The dimension of the secant variety of the matrix M representing the disjointness function satisfies $\dim(\sigma_2(V(M))) \geq n/2$ for all $n \times n$ 0-1 matrices M with $M[i][j] = 1$ iff $i \cap j = \emptyset$.

Rationale

Secant varieties capture the geometry of linear combinations of points, which mirrors the information-theoretic constraints in distributed computation. The dimension lower bound translates to communication complexity via the relation between algebraic degeneracy and protocol efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.25s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0be076e5.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0be076e5.py", line 53, in run_trial
    dim_secant_variety = secant_variety_dimension(M)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0be076e5.py", line 51, in secant_variet
    return rank(rank_2_flattening)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0be076e5.py", line 35, in rank
    rows[0], rows[i] = rows[i], rows[0]
                            ^
UnboundLocalError: cannot access local variable 'i' where it is not associated with a value
```

Judge reasoning

Test crashed due to unbound local variable error, preventing dimension calculation | next: Fix the rank computation code to handle row swaps properly and re-run experiments

---

Irreducible Component Count Bounds Circuit Complexity for CNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry × Boolean Circuits
- **Recorded**: 2026-05-01 02:01 UTC
- **Entry ID**: `1f65a5eb95d6`

Statement

For a random 3-CNF formula F with n variables, the number of irreducible components of the variety defined by its clauses is Θ(log s(F)), where s(F) is the minimal circuit size computing F.

Rationale

Irreducible components capture structural complexity of polynomial systems; their count may reflect the 'geometric complexity' of solving F, linking algebraic geometry to circuit size via the Nullstellensatz's degree bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6a3bc472.py", line 142, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6a3bc472.py", line 126, in run_trial
    component_count = irreducible_components(clauses, n)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6a3bc472.py", line 109, in irreducible_
    A[x][x + n + 1] += 1
    ~~~~^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before collecting data; cannot assess conjecture validity | next: Fix the index out of range error in the irreducible_components function and re-run tests

---

Schur-Weyl Multiplicity Gap in Permanent Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl duality × Communication matrix rank
- **Recorded**: 2026-05-01 02:29 UTC
- **Entry ID**: `39e42593cacd`

Statement

For random 3-CNF formulas with n variables, the multiplicity of the symmetric square representation in the decomposition of the permanent's communication matrix exceeds that of the determinant's matrix by Ω(2^n).

Rationale

Schur-Weyl duality decomposes tensor spaces into irreducible representations; permanent's higher multiplicity in symmetric squares could reflect its inherent symmetry complexity, creating a gap exploitable for communication lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.05s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with deterministic seed configuration

---

Matroid Rank and Minimal Circuit Size for Boolean Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Minimal Circuit Size (MCSP)
- **Recorded**: 2026-05-01 03:07 UTC
- **Entry ID**: `4d473e78494e`

Statement

For any boolean function f: {0,1}^n → {0,1}, the minimal circuit size of f is at least the rank of the matroid formed by the row space of its truth table over GF(2).

Rationale

The linear algebra structure of the truth table imposes constraints on the circuit complexity, suggesting that the rank of the matroid provides a lower bound on the minimal circuit size.

Novelty

- Judge: `INCONCLUSIVE` over 9 arXiv hits

Top hits consulted: - [2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits - [2003.09703v1] Variance function of boolean additive convolution - [1112.2313v2] QBF-Based Boolean Function Bi-Decomposition - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0

Empirical Test

- exit code: `124`, elapsed: `240.30s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing conclusive results; cannot determine support fraction or counterexample. | next: Increase timeout duration and re-run test with optimized code to handle larger instances

---

Bounded Arithmetic Proof Length and Circuit Complexity for Tautologies

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded Arithmetic (Cook-Reckhow-Krajicek) × Proof Length in Extended Frege System
- **Recorded**: 2026-05-01 03:27 UTC
- **Entry ID**: `94191613e108`

Statement

For any tautology in CNF with n variables, the minimal proof length in the Extended Frege system is Θ(circ_size), where circ_size is the minimal circuit size required to compute the tautology.

Rationale

Bounded arithmetic theories like S_{12} can simulate proof systems, and their proof complexity might be tied to the circuit complexity of the formulas they prove. By linking proof length to circuit size, we might uncover structural properties that separate bounded arithmetic theories.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_547f5a7e.py", line 99, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_547f5a7e.py", line 71, in run_trial
    cnf = generate_cnf(n)
          ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_547f5a7e.py", line 22, in generate_cnf
    cnf = [[random.choice([-i, i]) for _ in range(random.randint(1, n))] for _ in range(n)]
                              ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

Test crashed due to NameError before collecting data, preventing evaluation of support fraction or counterexamples. | next: Fix the NameError in generate_cnf by defining 'i' (likely a typo for 'n') and rerun tests with multiple seeds

---

Duality-Preserved Homotopy Stability

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC) × Homotopy-stable phase spaces
- **Recorded**: 2026-05-01 03:54 UTC
- **Entry ID**: `52b8188bdfa4`

Statement

If a tropical circuit C satisfies A3's homotopy-stability condition, then its duality-flipped counterpart C' also exhibits homotopy-stable phase spaces under weight perturbations.

Rationale

A3 asserts that provable $\forall\Sigma_1^b$ sentences correspond to homotopy-stable phase spaces. Duality Flip preserves functional equivalence on integer inputs (per operations), so stability under perturbations should carry over to the dual circuit, maintaining the connection between proof validity and phase robustness.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1207.2443v2] Tropical Teichmuller and Siegel spaces - [1204.6154v2] Local Tropicalization - [1204.3875v2] Tropicalizing vs Compactifying the Torelli morphism - [0706.3841v1] Arithmetic lattices and weak spectral geometry - [1512.00415v2] Étale homological stability and arithmetic statistics

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
e_holds': False, 'counterexample': 'Solution difference exceeds epsilon'}
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h
TRIAL: {'metric_name': 'homotopy_stability', 'metric_value': 0, 'instances_tested': 2, 'conjecture_h

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1d1fe44e.py", line 73, in <module>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1d1fe44e.py", line 73, in <genexpr>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                                   ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing required validation; counterexamples found but insufficient data to confirm falsification | next: Investigate crash cause in test_1d1fe44e.py's seed handling

---

Symmetric Power Partition Dominance in Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × 3-SAT Instances
- **Recorded**: 2026-05-01 04:01 UTC
- **Entry ID**: `567149797f34`

Statement

For random 3-SAT instances with n variables, the number of times the partition (n) appears in the symmetric power decomposition of the permanent polynomial exceeds that in the determinant polynomial by $\Omega(\sqrt{n})$ for all $n \leq 40$.

Rationale

Schur-Weyl duality decomposes tensor powers into irreps; permanent's symmetric nature creates more dominant partitions than determinant's alternating structure, revealing algebraic asymmetry central to GCT's permanent vs determinant separation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ecd9910.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ecd9910.py", line 86, in run_trial
    instance = random_3sat_instance(n, m)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ecd9910.py", line 22, in random_3sat_i
    clause = random.sample(variables, 3)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

Test crashed due to invalid population type in random sampling, preventing data collection. | next: Fix the population sequence issue in random_3sat_instance by converting variables to a list

---

Schur-Weyl Multiplicity Gap in Permanent vs Determinant Tensor Powers

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Permanent Polynomial
- **Recorded**: 2026-05-01 04:36 UTC
- **Entry ID**: `4b47ab077abe`

Statement

For all $n \geq 2$, the multiplicity of the trivial representation in the decomposition of S^k(perm_n) exceeds that in S^k(det_m) by a factor of 2^{Ω(n)} for m = n^{1.5} and k = log n.

Rationale

Schur-Weyl duality decomposes symmetric powers into irreducible representations, with trivial representation multiplicities reflecting symmetry-breaking complexity. A gap here could indicate structural separation between permanent and determinant orbit closures under linear substitutions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `5.89s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ff917c49.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ff917c49.py", line 97, in run_trial
    trivial_perm = trivial_representation_multiplicity(perm_poly)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ff917c49.py", line 64, in trivial_repre
    return gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ff917c49.py", line 46, in gaussian_elim
    n = len(A)
        ^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

Test crashed due to TypeError in Gaussian elimination, preventing data collection | next: Fix the gaussian_elimination function to handle integer inputs properly

---

Bounded Arithmetic Proof Length and 3-SAT Tautology Resolution Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded Arithmetic × Resolution Proof Length
- **Recorded**: 2026-05-01 05:04 UTC
- **Entry ID**: `cd7f4a991a5d`

Statement

For a random 3-CNF tautology with n variables, the length of the proof in the bounded arithmetic system P^0 is $\Theta(n^2)$, where n is the number of variables.

Rationale

Bounded arithmetic systems can model polynomial-time reasoning, and their proof lengths may reflect the inherent complexity of tautologies. Resolution proof length is a well-studied measure of complexity, making this a natural bridge.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
njecture_holds': False, 'counterexample': 'no_valid_tautologies'}
TRIAL: {'seed': 547, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 593, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 631, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 677, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 727, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 773, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 821, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 877, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test
TRIAL: {'seed': 929, 'metric_name': 'resolution_proof_length', 'metric_value': None, 'instances_test

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1a62a63.py", line 73, in <module>
    mean_length = sum(r['metric_value'] for r in results) / len(results)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: unsupported operand type(s) for +: 'int' and 'NoneType'
```

Judge reasoning

Test crashed due to TypeError from None metric values; results cannot be trusted | next: Fix test to handle None metric values and rerun with proper validation

---

Noncommutative L^p Norm Lower Bounds for Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-01 05:36 UTC
- **Entry ID**: `bd540f51275f`

Statement

For any n×n matrix M defining a Disjointness instance, the noncommutative L^p norm ||M||_p ≥ Ω(n^{1-1/p}) for p ∈ (1,2]. This lower bound matches the known Ω(n) communication complexity for Disjointness.

Rationale

Noncommutative L^p norms capture the geometry of matrix operators in noncommutative spaces, which may reveal intrinsic dimensionality constraints mirroring communication complexity lower bounds. The p=2 case relates to operator norms, while intermediate p values could expose hidden structure in tensor products.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.39s`


```
7.256808892482084, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 28.966628278922446, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 30.49390723052147, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 31.139511681667063, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 34.30918248997468, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 26.052175797574254, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 28.217523437965372, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 29.551246746775025, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 30.34100726290498, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 23.24935058115917, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 26.031519958125916, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 27.67917169898806, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 27.558827236719782, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'lp_ratio', 'metric_value': 27.538744286264862, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=29.0842492209495 std=0.0 support_fraction=1.0
```


Judge reasoning

Safety rail: critic_challenge | original judge: All 13 trials (100%) support the conjecture with mean lp_ratio ≈29.08, exceeding the required 80% support threshold. | next: Test larger n×n matrices to verify scalability of the lower bound

---

Phase Merging Complexity Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Circuit Weight Analysis (TCWA) in BOUNDED_ARITHMETIC × Minkowski Sum Complexity
- **Recorded**: 2026-05-01 05:58 UTC
- **Entry ID**: `3b94aaddf6cd`

Statement

The tropical proof rank of a phase space obtained via Phase Merging of two circuits is at most the sum of their individual tropical proof ranks.

Rationale

Tests A1 (Phase Stability Axiom) by verifying if merged phase spaces' complexity scales additively with individual ranks, ensuring bounded arithmetic proof strength remains decomposable under superposition.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e5df8221.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e5df8221.py", line 68, in run_trial
    r1 = tropical_rank(A)
         ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e5df8221.py", line 60, in tropical_rank
    x = gaussian_elimination(A, b, 2)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e5df8221.py", line 42, in gaussian_elim
    factor = (A[i][i] * mod_inverse(A[i][i], p)) % p
                        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e5df8221.py", line 29, in mod_inverse
    raise ValueError("Modular inverse does not exist")
ValueError: Modular inverse does not exist
```

Judge reasoning

Test crashed due to modular inverse error, preventing validation of conjecture | next: Check if modulus is prime and verify matrix entries' coprimality with modulus

---

Noncommutative L^p Norm Lower Bounds for Disjointness

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-01 06:07 UTC
- **Entry ID**: `0acf2c6083e2`

Statement

For any n×n matrix M representing the Disjointness function, the noncommutative L^p norm ||M||_p ≥ Ω(√n) for p = log n. This bound is tight up to polylog(n) factors.

Rationale

Noncommutative L^p norms capture operator-space geometry of matrices, which could expose hidden structure in communication protocols. The log n choice of p aligns with known lower bounds for multi-party protocols, while avoiding standard operator norms that have been extensively studied.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
nces_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.141263493487763, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 16.677331100759226, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.30487471363759, 'instances_teste
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 16.988451401307042, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.164707688084967, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 15.695162405340357, 'instances_tes
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TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.581092996066843, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tes
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.164707688084967, 'instances_tes
RESULT: SUPPORTED mean=17.1583 std=0.6962 support_fraction=1.00
```

Judge reasoning

All trials used n=1, preventing scaling analysis. Metric values show no growth with n, contradicting $\Omega(\sqrt{n})$ claim. Statistical significance invalid with 1 instance per trial. | next: Test with varying n ≥ 1000 to observe scaling behavior of noncommutative L^p norm

---

Real Rank of Coefficient Matrices for AC^0 PARITY Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuits Computing PARITY
- **Recorded**: 2026-05-01 06:41 UTC
- **Entry ID**: `80c8560ef86e`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the real rank of the matrix formed by the coefficients of the linear terms in C is $\Omega(\log \text{size}(C))$

Rationale

The real rank captures geometric constraints on linear representations, and AC^0 circuits must exhibit high-rank coefficient matrices to approximate PARITY, which is inherently linear.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.64s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with identical parameters

---

Tropical Circuit Homotopy Stability under Weight Accumulation

- **Verdict**: `BARRIER_HIT`
- **Bridge**: {'framework_name': 'Tropical Circuit Weight Analysis (TCWA)', 'math_branch': 'BOUNDED_ARITHMETIC'} × {'complexity_theoretic_object': 'Tropical Circuit Model'}
- **Recorded**: 2026-05-01 07:13 UTC
- **Entry ID**: `d2641eee1d4e`

Statement

For any tropical circuit C and input x, the weight accumulation trace of C under Circuit Homotopy deformation is bounded by the Tropical Proof Rank of the corresponding bounded arithmetic theory, thereby preserving the homotopy-stable phase space under perturbation.

Rationale

This sub-conjecture tests Axiom A3 (Tropical Soundness) by examining the relationship between the weight accumulation trace and the Tropical Proof Rank. It should hold because the Tropical Proof Rank is designed to bound the minimum number of phase transitions required to simulate a bounded arithmetic proof line, and the weight accumulation trace is a measure of the circuit's behavior under deformation. If the weight accumulation trace is bounded by the Tropical Proof Rank, it would provide evidence for the tropical soundness of the circuit model.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Orbit Closure Dimension vs. Boolean Circuit Size

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Geometric Complexity Theory × Boolean Circuits
- **Recorded**: 2026-05-01 07:17 UTC
- **Entry ID**: `521f38945837`

Statement

For any Boolean function f, the dimension of the orbit closure of the determinant polynomial under the action of the general linear group is upper bounded by the size of the smallest Boolean circuit computing f. Specifically, for a Boolean circuit C of size s, the dimension of the orbit closure of the determinant polynomial is at most O(s^2).

Rationale

The orbit closure dimension is a measure of the complexity of a polynomial under the action of a group, and the determinant polynomial is closely related to the permanent polynomial, which is a key object of study in geometric complexity theory. By relating the orbit closure dimension to Boolean circuit size, we may be able to leverage the power of geometric complexity theory to prove lower bounds on Boolean circuit size.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Homotopy Type and AC^0 Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Homotopy Theory × AC^0 Circuit Size
- **Recorded**: 2026-05-01 07:55 UTC
- **Entry ID**: `5cbad98886db`

Statement

For any Boolean function f, the homotopy type of its decision tree is correlated with the size of its AC^0 circuit. Specifically, if the homotopy type of the decision tree has a non-trivial fundamental group, then the AC^0 circuit size is at least Ω(n^2).

Rationale

Homotopy theory provides a way to study the topological structure of decision trees, which can be used to understand the complexity of Boolean functions. The fundamental group of the decision tree captures the ‘holes’ in the tree, which can be related to the complexity of the function. The correlation between homotopy type and AC^0 circuit size can be used to establish lower bounds on the circuit size.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [1003.3820v1] Double groupoids and homotopy 2-types - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [2003.09703v1] Variance function of boolean additive convolution

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
unterexample": ""}
TRIAL: {"seed": 389, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 421, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 463, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 503, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 547, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 593, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 631, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 677, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 727, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 773, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 821, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 877, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
TRIAL: {"seed": 929, "metric_name": "AC^0 circuit size", "metric_value": 32, "instances_tested": 1, "c
RESULT: SUPPORTED mean=32.0 std=0.0 support_fraction=1.0
```


Judge reasoning

The critic challenges the result due to a small sample size and potential 'n too small' failure mode, and the pre-registered support condition is not unambiguously met. | next: Increase the sample size and test with larger values of n to verify the conjecture's scalability

---

Littlewood-Richardson Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl duality × Homogeneous polynomial complexity
- **Recorded**: 2026-05-01 08:38 UTC
- **Entry ID**: `52ba9a0b4010`

Statement

For all n ≥ 2, the sum of Littlewood-Richardson coefficients for the decomposition of perm_n^⊗k into irreps λ ⊢ n^2 has Θ(n^{k-1}) growth, while det_m^⊗k has Θ(n^{k}) growth for m = ⌊n^{1.5}⌋.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations, revealing structural asymmetries between permanent and determinant. The conjecture posits that the growth rate of specific coefficient sums reflects inherent complexity gaps detectable via representation-theoretic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.54s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing verification of growth rates. Pre-registered support condition cannot be evaluated without data. | next: Optimize/test with higher time limits and larger n/k values to resolve timeout

---

Algebraic Curve Genus and Communication Complexity Lower Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry × Communication Complexity
- **Recorded**: 2026-05-01 09:12 UTC
- **Entry ID**: `e7614c62651c`

Statement

For any Boolean function f: `{0,1}^n × {0,1}^n → {0,1}`, the genus g of the algebraic curve defined by its communication matrix satisfies g ≥ log^2(Comm(f))

Rationale

Algebraic curves encode structural constraints on polynomial representations of communication matrices. The genus, a topological invariant, may capture inherent complexity barriers through its relation to polynomial degree and ramification, offering a new lens to analyze communication complexity lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.60s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug crash cause

---

Matroid Girth of Monotone DNF Implies k-CLIQUE Circuit Size Lower Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-01 09:26 UTC
- **Entry ID**: `c18d42c448ab`

Statement

Let F be a monotone DNF formula representing the k-CLIQUE function on n-vertex graphs. Let M(F) be the simple binary matroid whose ground set is the set of terms of F, and whose independent sets are subsets of pairwise edge-disjoint terms. Then the girth of M(F) — the size of the smallest circuit (minimal dependent set) in M(F) — is at least k. Furthermore, if a monotone DNF has matroid girth < k, it cannot compute k-CLIQUE on all n-vertex graphs for n ≥ 3k.

Rationale

The girth in this matroid measures the smallest number of edge-disjoint but combinatorially overlapping terms required to create a dependency, which corresponds to unavoidable overcounting in clique detection. High girth forces structural complexity in the DNF that reflects the necessity of large support for covering all k-cliques without false positives. This connects extremal combinatorics in matroids to proof complexity in monotone circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.13s`

```
: 'DNF with girth < 3'}
TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
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TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
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TRIAL: {'metric_name': 'Matroid Girth', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds'
RESULT: INCONCLUSIVE insufficient data
```

Judge reasoning

Test instances use n=1 (too small) and matroid girth ≥2 is trivial, making the conjecture's bound vacuous. Counterexample is minimal girth, not violating conjecture's conditions. | next: Test with n ≥ 3k for k=3 (n≥9) and verify girth ≥3 for k-CLIQUE DNFs

---

Phase Merging Additivity in Tropical Proof Rank

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Circuit Weight Analysis (TCWA) in BOUNDED_ARITHMETIC × Merged Phase Space Complexity
- **Recorded**: 2026-05-01 09:35 UTC
- **Entry ID**: `e9b2b689bd8f`

Statement

The tropical proof rank of a merged phase space produced by Phase Merging of two circuits C_1 and C_2 is at most the sum of their individual tropical proof ranks.

Rationale

This tests A1 (Phase Stability Axiom) by examining how phase space refinement under superposition scales with proof complexity. If merging two circuits with ranks r_1 and r_2 yields a merged rank ≤ $r_1 + r_2$, it validates that phase transitions remain polynomially bounded under composition, crucial for separating bounded arithmetic theories.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
ure_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
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TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
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TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
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TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'tropical_proof_rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
RESULT: SUPPORTED mean=5.0 std=0.0 support_fraction=1.0
```

Judge reasoning

While 100% support is observed, the critic highlights insufficient instance diversity (n=1) and metric saturation (fixed value 5) that fail to rigorously validate the conjectured property about phase merging's impact on tropical proof rank. | next: Test phase merging scenarios with varying circuit structures and measure tropical proof rank evolution to challenge metric saturation

---

Spectral Gap Lower Bounds for Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length
- **Recorded**: 2026-05-01 09:55 UTC
- **Entry ID**: `58628c03994f`

Statement

For any connected graph G with n vertices, the Resolution proof length of its Tseitin formula is ≥ 2^{Ω(λ_2)}, where λ_2 is the second smallest eigenvalue of G's Laplacian matrix. For non-expanders (λ_2 ≤ 1/√n), this bound is vacuous.

Rationale

The spectral gap λ_2 governs expansion properties critical for Tseitin formulas' hardness. Larger λ_2 implies stronger expansion, which amplifies the need for exponential proof size in Resolution. This connects algebraic structure to proof complexity via eigenvalue analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
ossible'}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': None, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Resolution length', 'metric_value': None, 'instances_tested': 1, 'conjecture_
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TRIAL: {'metric_name': 'Resolution length', 'metric_value': None, 'instances_tested': 1, 'conjecture_
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TRIAL: {'metric_name': 'Resolution length', 'metric_value': None, 'instances_tested': 1, 'conjecture_

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_be5de29e.py", line 106, in <module>
    mean = sum(results) / len(results)
           ^^^^^^^^^^^^
TypeError: unsupported operand type(s) for +: 'int' and 'NoneType'
```

Judge reasoning

Test crashed before producing valid data, preventing evaluation of conjecture support/falsification. | next: Fix the test code to handle missing metric values and re-run experiments

---

Permutation Polynomial Degree Bounds ABP Size for NC¹ Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Permutation Polynomials over Finite Fields × Algebraic Branching Programs (ABPs)
- **Recorded**: 2026-05-01 10:34 UTC
- **Entry ID**: `f56efd7da9d8`

Statement

For every Boolean function f: {0,1}^n → {0,1} in NC¹, the minimal degree of a permutation polynomial over GF(2^n) representing f is at most the size of the smallest ABP computing f. Conversely, if an ABP has size s, then the minimal degree of such a polynomial is Ω(s / log n).

Rationale

Permutation polynomials over finite fields capture symmetry in Boolean functions, while ABP size reflects algebraic complexity. Linking their growth rates could reveal structural constraints on NC^1 computations via algebraic representations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.29s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of conjecture. | next: Increase timeout duration and re-run test with additional debugging instrumentation

---

SOS Moment Matrix Rank and Max-CUT Approximation Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex Geometry × Sum-of-Squares Degree of Max-CUT
- **Recorded**: 2026-05-01 11:29 UTC
- **Entry ID**: `c27792e507e5`

Statement

For every n-vertex Max-CUT instance, the rank of its degree-d SOS moment matrix is at least Ω(n^{1-1/d}) if the approximation ratio is below 0.878 - ε for any ε > 0.

Rationale

The rank of the SOS moment matrix encodes the geometry of the feasible region for semidefinite relaxations. High rank implies the relaxation cannot be captured by low-degree polynomials, directly linking geometric complexity to approximation guarantees.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9699fbd.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9699fbd.py", line 59, in run_trial
    M = degree_d_sos_moment_matrix(G, d)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9699fbd.py", line 31, in degree_d_sos_
```

```
    M[i * n**(k-1):(i+1)*n**(k-1), j*n**(k-1):(j+1)*n**(k-1)] += [[G[i][j]] * n**(2*k-2)]
    ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: list indices must be integers or slices, not tuple
```

Judge reasoning

Test crashed due to a TypeError, preventing data collection. Pre-registered support condition cannot be evaluated without successful runs. | next: Fix the TypeError in degree_d_sos_moment_matrix by correcting the list indexing operation

---

Schur-Weyl Multiplicity Gap in Permanent vs Determinant Tensor Powers

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × 3-SAT Instances
- **Recorded**: 2026-05-01 12:03 UTC
- **Entry ID**: `27563135d273`

Statement

For any 3-SAT instance with n variables, the multiplicity of the irreducible representation λ = (n−1,1) in the decomposition of the permanent polynomial's tensor power P^⊗k is strictly greater than that in the determinant polynomial's tensor power D^⊗k for all k ≥ 2, where P and D are homogeneous of degree n.

Rationale

Schur-Weyl duality links symmetric group representations to polynomial invariants. The permanent's symmetric nature under row/column permutations creates higher multiplicities for certain Young diagrams compared to the determinant's antisymmetric structure, potentially exposing algebraic barriers to efficient computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6009679.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6009679.py", line 51, in run_trial
    P_k = multiplicity(n - 1, 2)
          ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6009679.py", line 48, in multiplicity
    return plethysm_coefficient(lambda_, poly_tensor_power)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6009679.py", line 44, in plethysm_coef
    coeff += ((-1) ** (n - m)) * hook_length_formula(m) ** k
                                 ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6009679.py", line 39, in hook_length_f
    return math.factorial(n) / total
```

```
          ~~~~~~~~~~~~~~~~~^~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity. | next: Fix hook_length_formula implementation to handle edge cases in partition calculations

---

Multiplicity of Symmetric Powers in Permanent Tensor Decomposition

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Algebraic Groups × Boolean Circuit Complexity of Permanent
- **Recorded**: 2026-05-01 13:05 UTC
- **Entry ID**: `d4ccd48216c9`

Statement

For every n ≥ 2, the multiplicity of the trivial representation in the decomposition of the symmetric power S^n(perm_n) is Θ(n^2), where perm_n is the permanent polynomial of size n. This multiplicity equals the minimal number of gates in a depth-3 circuit computing perm_n over GF(2).

Rationale

Symmetric power decompositions encode how tensor products of polynomials interact with algebraic group actions. The permanent's asymmetric representation-theoretic structure could reveal inherent complexity barriers via nontrivial multiplicities, linking geometric invariants to circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.46s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing verification of conjecture. | next: Run test with extended time limit and higher resource allocation

---

Diophantine Solution Count Bounds Resolution Length in 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine Approximation × Resolution Proof Length
- **Recorded**: 2026-05-01 14:32 UTC
- **Entry ID**: `00e75e5287ed`

Statement

For a 3-SAT instance with n variables, the minimal resolution proof length is at most the number of integer solutions to the system S derived from the instance, where S consists of equations enforcing clause truth via linear Diophantine constraints modulo a prime $p \geq n+1$.

Rationale

Diophantine equations capture combinatorial constraints, and their solution counts may reflect the complexity of resolving logical contradictions. Modular arithmetic introduces structure that could correlate with proof complexity via bounded arithmetic frameworks.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6d675fda.py", line 125, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6d675fda.py", line 64, in run_trial
    clauses = [[random.choice([-i, i]) for _ in range(3)] for _ in range(n)]
                                ^
UnboundLocalError: cannot access local variable 'i' where it is not associated with a value
```

Judge reasoning

Test crashed due to unbound local variable ‘i’, preventing data collection | next: Fix the code error in line 64 where ‘i’ is used without definition

---

Singular Locus Dimension Bounds SOS Refutation Size for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Singularity Theory × SOS Refutation Size for 3-SAT
- **Recorded**: 2026-05-01 15:02 UTC
- **Entry ID**: `160abd91b150`

Statement

For a 3-SAT instance Φ with n variables, the dimension of the singular locus of the variety $V(\Phi)$ defined by its polynomial system satisfies $\dim(\mathrm{sing}(V(\Phi))) \geq \log_2(\mathrm{sos\_refutation\_size}(\Phi)) - 3n^2$. This holds for all Φ with at least one solution.

Rationale

Singular loci encode geometric obstructions to polynomial system solvability. SOS refutations leverage semidefinite programming to certify unsatisfiability, with complexity tied to the geometry of the feasible region. Singularities may create bottlenecks in the SOS hierarchy’s ability to capture global structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1141eb39.py", line 67, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1141eb39.py", line 43, in run_trial
    polynomials = [polynomial_from_clause(clause) for clause in instance]
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1141eb39.py", line 30, in polynomial_fr
    poly = sum(x_vars[var] if coeff == 1 else -x_vars[var] for coeff, var in zip([1, 1, 1], range(1, 4)))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1141eb39.py", line 30, in <genexpr>
    poly = sum(x_vars[var] if coeff == 1 else -x_vars[var] for coeff, var in zip([1, 1, 1], range(1, 4)))
               ~~~~~~^^^^^
KeyError: 1
```

Judge reasoning

Test crashed with KeyError during polynomial generation, preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Fix variable indexing in polynomial_from_clause to handle clause-to-polynomial conversion correctly

---

Expander Modulus Bounds Tseitin Proof Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Expander Graphs and Ramanujan Complexes × Tseitin Formula Proof Complexity
- **Recorded**: 2026-05-01 16:11 UTC
- **Entry ID**: `b0314b45545a`

Statement

For every Tseitin formula φ constructed from an expander graph G = (V, E) with n vertices and second eigenvalue λ, the proof complexity of φ in the Extended Frege system is at least Ω(n / λ^2).

Rationale

The connection between expander graphs and proof complexity has been explored in various contexts. The use of Ramanujan complexes, which generalize expander graphs to higher dimensions, might provide a new perspective on the proof complexity of Tseitin formulas. The second eigenvalue λ of the graph is a measure of its expansion, and it is reasonable to expect that this affects the difficulty of proving Tseitin formulas constructed from the graph.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.02s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_409ff85c.py", line 65, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_409ff85c.py", line 49, in run_trial
    λ = compute_second_eigenvalue(A)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_409ff85c.py", line 34, in compute_secon
    v /= math.sqrt(sum(x * x for x in v))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

The test crashed before producing data, making it impossible to evaluate the conjecture. | next: Fix the bug in the compute_second_eigenvalue function to allow the test to run to completion.

Tropical Circuit Weight Profile Complexity Bound

- **Verdict**: INCONCLUSIVE
- **Bridge**: {'framework_name': 'Tropical Circuit Weight Analysis (TCWA)', 'math_branch': 'BOUNDED_ARITHMETIC'} × {'complexity_theoretic_object': 'Weight Profile'}
- **Recorded**: 2026-05-01 16:36 UTC
- **Entry ID**: 7e57991f294f

Statement

The weight profile of a tropical circuit for a formula encoding a proof line in $I\Delta_0$ has a length bounded by the tropical proof rank of the formula, under uniform input scaling.

Rationale

This sub-conjecture tests Axiom A1 (Phase Stability Axiom) by exploring the relationship between the tropical proof rank and the weight profile complexity. If the weight profile length is bounded by the tropical proof rank, it would provide evidence for the phase stability of tropical circuits under uniform input scaling, which is a key aspect of Axiom A1.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1503.01170] Integer Addition and Hamming Weight - [s2:5d5145ad0400a8508cf97f469 Automata, Languages and Programming: 19th International Colloquium, Wien, Austria, July 13-17, 1992. Proceedings

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ee3ff6.py", line 50, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ee3ff6.py", line 41, in run_trial
   "counterexample": f"Formula: {formula}, Circuit: {circuit}, Weight Profile Length: {weight_profile

NameError: name 'conjecture_holds' is not defined
```

Judge reasoning

Test crashed before producing data; cannot evaluate conjecture support or falsification. | next: Fix the test code to handle NameErrors and re-run with multiple seeds

---

Finite Geometry Rank and ACC^0 Circuit Size for 3-SAT Instances

- **Verdict**: INCONCLUSIVE
- **Bridge**: Finite Geometry × Boolean Circuit Size (ACC^0)
- **Recorded**: 2026-05-01 16:47 UTC
- **Entry ID**: 8255cc53c91e

Statement

For any 3-SAT instance Φ with n variables, the rank of its incidence matrix over GF(2) is Θ(log n) if and only if Φ can be computed by an ACC^0 circuit of size O(n^ε) for some ε < 1.

Rationale

The incidence matrix's rank over GF(2) captures linear dependencies among clauses, which may constrain ACC^0 circuits' ability to simulate Φ. High rank implies complex dependencies, requiring larger circuits, while low rank suggests structured dependencies amenable to small ACC^0 implementations.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 60, in run_trial
    rank = gaussian_elimination(incidence_matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 32, in gaussian_elim
    matrix[j][k] -= factor * matrix[i][k]
    ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during Gaussian elimination, preventing data collection. Support condition cannot be evaluated without successful runs. | next: Fix the Gaussian elimination implementation to handle matrix dimensions properly and re-run tests

---

Moment Matrix Eigenvalue Gap and SOS Degree for Max-CUT

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Real Algebraic Geometry × SOS Degree of Max-CUT
- **Recorded**: 2026-05-01 17:19 UTC
- **Entry ID**: `fa0b65fa4701`

Statement

For any Max-CUT instance with n variables, the minimal eigenvalue λ_min of its degree-d SOS moment matrix satisfies $\lambda\_min \geq \Omega(1/n^2)$. Thus, achieving a 0.879-approximation requires $d = \Omega(n)$.

Rationale

The SOS moment matrix's eigenvalue structure encodes the feasibility of polynomial certificates for Max-CUT. A lower bound on λ_min implies the hierarchy must scale with n to avoid degenerate certificates, linking algebraic geometry to SOS complexity.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Additive Energy Threshold and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-01 18:20 UTC
- **Entry ID**: `cefe1013b737`

Statement

For a Boolean function f on n variables, if its additive energy E(f) ≥ n^{2 - ε}, then the minimal ACC^0 circuit size of f is at least Ω(n^{1+ε}) for any ε > 0.

Rationale

High additive energy implies the function is structured in a way that resists efficient computation by ACC^0 circuits, which are limited in their ability to handle structured functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.37s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug crash logs

---

Tropical Derivation Depth Reflects Proof Rank in Bounded Arithmetic

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Circuit Weight Analysis (TCWA) — Bounded Arithmetic × circuit depth (in AC^0)
- **Recorded**: 2026-05-01 19:17 UTC
- **Entry ID**: `cb8a67675b12`

Statement

For any tropical circuit C computing a function provably total in $I\Delta_0$, the maximum depth of the tropical derivation δ_C(x) over all inputs x ∈ {0,1}^n is at least the Tropical Proof Rank of the corresponding proof line in V^0.

Rationale

This sub-conjecture tests Axiom A1 (Phase Stability Axiom) by linking syntactic proof complexity — as captured by Tropical Proof Rank — to a structural property of derivations. The depth of the tropical derivation reflects how many sequential phase-sensitive decisions are required during evaluation. Since $I\Delta_0$-provably total functions are expected to have limited combinatorial complexity, their derivations should not require deep chains of active path selections unless simulating nontrivial proof structure. Thus, derivation depth should grow at least as fast as the minimal number of phase transitions needed to simulate a proof line.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
(no output)
```

Judge reasoning

The critic challenges the validity of small n tests where depth scales trivially with n, violating the ‘n too small’ failure mode. Empirical support fraction is undefined due to empty per_seed_brief. | next: Test with $n \geq 20$ to observe non-trivial depth scaling and validate the conjecture’s metric relevance